

# JEE MAIN REVISION MASTER: CIRCLE 2026 1

## QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

### 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                 | Frequency | Category           |
|---------------|----------------------------------------------|-----------|--------------------|
| Moderate      | Circle Equation, Tangents & Chord of Contact | 3 PYQs    | Circles            |
| Standard      | Tangents, Normals & Orthogonal Circles       | 0 PYQs    | Tangents & Circles |

### 2. Master Formula Reference Card

| Concept / Topic                              | Mathematical Formulation & Governing Equations                                                                                                                       |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Circle Equation, Tangents & Chord of Contact | $r = \sqrt{g^2 + f^2 - c}$ , $\text{Tangent: } y = mx \pm a\sqrt{1 + m^2}$ , $\text{Chord of contact: } T = 0$ , $\text{Orthogonal: } 2g_1g_2 + 2f_1f_2 = c_1 + c_2$ |
| Tangents, Normals & Orthogonal Circles       | $\text{Tangent: } y = mx \pm a\sqrt{1 + m^2}$ , $\text{Length of tangent: } L = \sqrt{S_1}$ , $\text{Orthogonal: } 2g_1g_2 + 2f_1f_2 = c_1 + c_2$                    |

### 3. Deep-Dive Concepts & Linked PYQ Evidence

#### 3.1 Circle Equation, Tangents & Chord of Contact (Moderate Yield — 3 PYQs)

**Core Principle:** Standard and general equations of circles, condition for line  $y = mx + c$  to be tangent, and director circle  $x^2 + y^2 = 2a^2$ .

■ **Caution / JEE Pitfall:** The center of  $x^2 + y^2 + 2gx + 2fy + c = 0$  is  $(-g, -f)$ , NOT  $(+g, +f)$ .

■ **Speed Trick:** The locus of the point of intersection of two perpendicular tangents to a circle is its director circle (radius  $\sqrt{2}a$ ).

**Representative Source PYQs:**

- [2026 Shift] Q#1 (MCQ): "The area enclosed by the curve  $x = \sqrt{3}\cos\theta, y = \sqrt{3}\sin\theta$  is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "If in two circles, arcs of the same length subtend angles  $30^\circ$  and  $78^\circ$  at the centre, then the ratio of their radii is..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "The maximum area of a rectangle inscribed in the circle  $(x+1)^2 + (y-3)^2 = 64$  is..." [Ans: N/A]

### 3.2 Tangents, Normals & Orthogonal Circles (Standard Yield — 0 PYQs)

**Core Principle:** Point and slope forms of circle tangents, director circle  $x^2 + y^2 = 2a^2$ , and orthogonality condition.

■ **Caution / JEE Pitfall:** The normal to a circle at any point ALWAYS passes through the center of the circle  $(-g, -f)$ .

■ **Speed Trick:** Common chord of two intersecting circles  $S_1 = 0$  and  $S_2 = 0$  is given directly by the linear equation  $S_1 - S_2 = 0$ .

**Representative Source PYQs:**

- **[2026 Shift] Q#1 (MCQ):** "The area enclosed by the curve  $x = \sqrt{3}\cos\theta, y = \sqrt{3}\sin\theta$  is..." [Ans: N/A]
- **[2024 Shift] Q#2 (MCQ):** "If in two circles, arcs of the same length subtend angles  $30^\circ$  and  $78^\circ$  at the centre, then the ratio of their radii is..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

The area enclosed by the curve  $x = \sqrt{3}\cos\theta, y = \sqrt{3}\sin\theta$  is

- (A)  $\sqrt{3}\pi$  sq. units
- (B)  $9\pi$  sq. units
- (C)  $6\pi$  sq. units
- (D)  $3\pi$  sq. units

**Correct Answer: None**

**Solution Analysis:** Given,  $x = \sqrt{3}\cos\theta, y = \sqrt{3}\sin\theta$  We know the standard parametric form of a circle is  $x = a\cos\theta, y = a\sin\theta$ . This represents a circle with centre at the origin and radius  $a$ . Here,  $a = \sqrt{3}$ . So the radius of the circle is  $\sqrt{3}$ . Area enclosed by a circle is  $\pi r^2$ . So,  $\text{Area} = \pi(\sqrt{3})^2$ .  $\text{Area} = 3\pi$ . Hence, the area enclosed by the curve is  $3\pi$  sq. units. Correct option: Option D.